Introduction of Dermatology

Prof. Amira Ali Abdel Motaleb

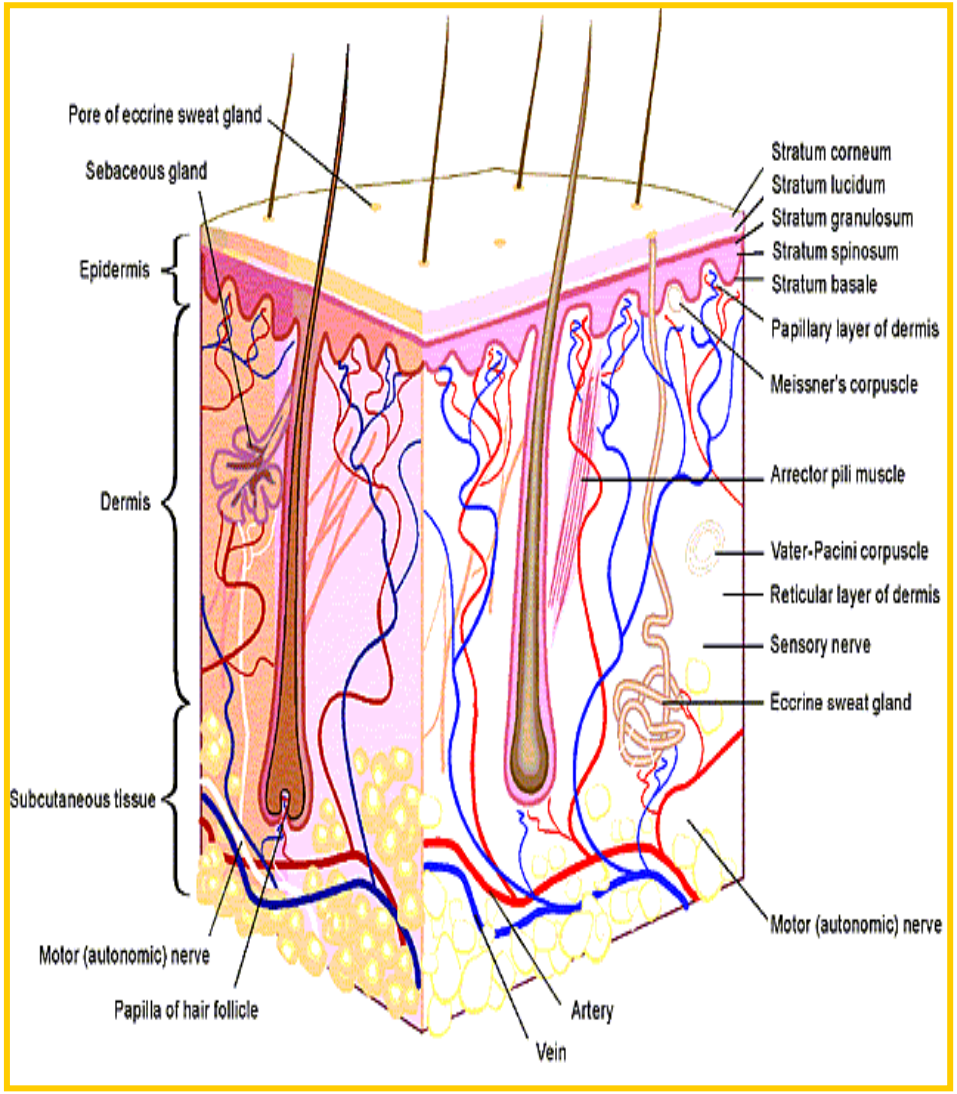
Prof. of Dermatology Venereology and Andrology ,
Assuit Universty

Structure of the skin

Skin

- **Skin is the largest organ of the body.**

- **The skin is composed of 3 layers: epidermis, dermis and subcutaneous tissue.**



Epidermis

- The adult epidermis is composed of 3 basic cell types: keratinocytes, melanocytes, and Langerhans cells.

- The *keratinocyte* is the principal cell of the epidermis.

- The epidermis is divided into 4 main layers beginning with the innermost layer (St. basal, spinosum, granulosum, corneum).

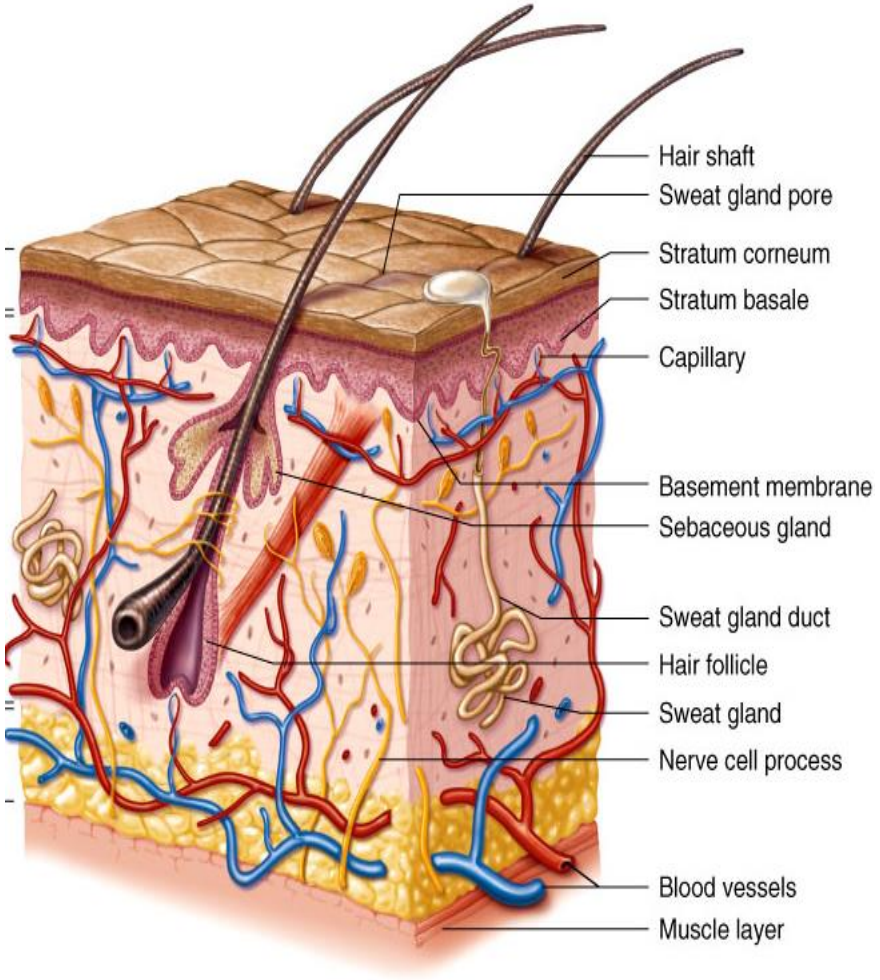
- The *Langerhans* cell is a dendritic cell present in the squamous cell layer (Stratum spinosum).

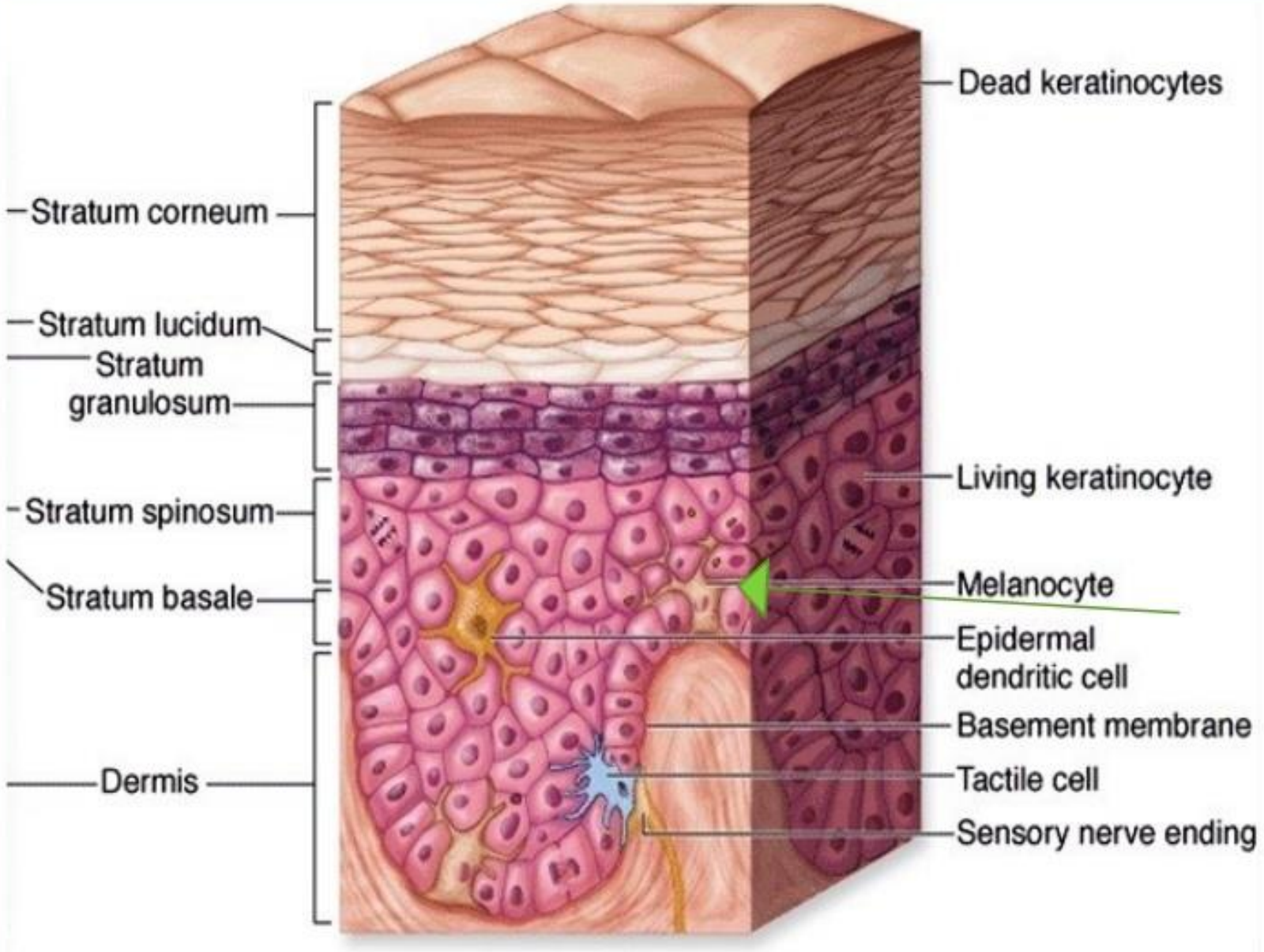
- Functionally, it is of the monocyte-macrophage lineage and originates in bone marrow.

- Langerhans cell plays a role in some immunological reactions.

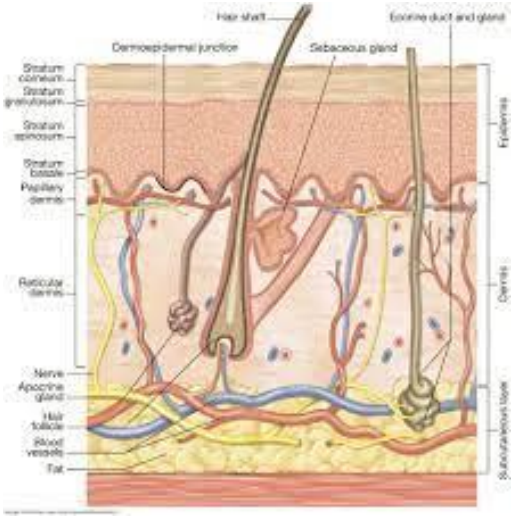
- *The melanocytes* are dendritic cells produce pigment and found in the basal layer

- Epidermal appendages include: hair, nails, sebaceous glands and sweat glands



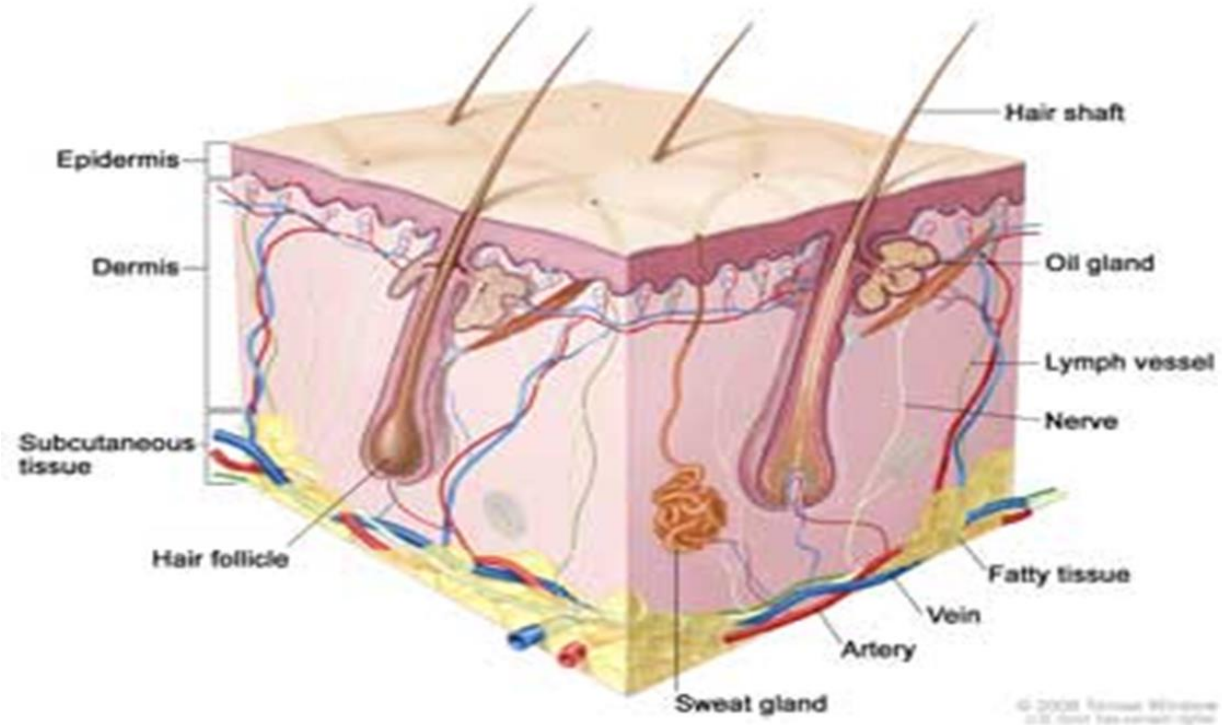

Dead keratinocytes
Stratum corneum
Stratum lucidum
Stratum granulosum
Stratum spinosum
Stratum basale
Dermis
Living keratinocyte
Melanocyte
Epidermal dendritic cell
Basement membrane
Tactile cell
Sensory nerve ending

Dermis

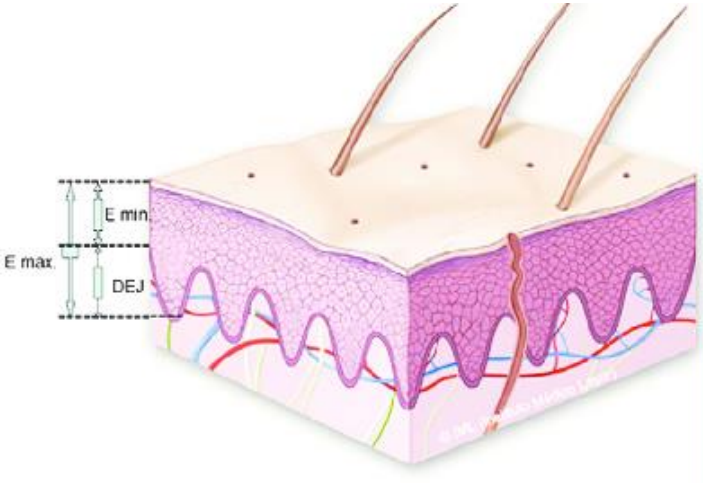


- It is bounded externally by its junction with the *epidermis* and internally by *subcutaneous fat*.
- It contributes to 15-20% of the total weight of the human body.
- It consists of upper papillary dermis and lower reticular dermis
- It largely consists of a supporting matrix or ground substance and connective tissue formed mainly of collagen and elastin.

- The predominant cellular elements in the dermis are fibroblasts.
- The dermis also contains blood vessels, lymphatics, as well as peripheral, sensory and motor nerve endings.



Dermo- epidermal Junction



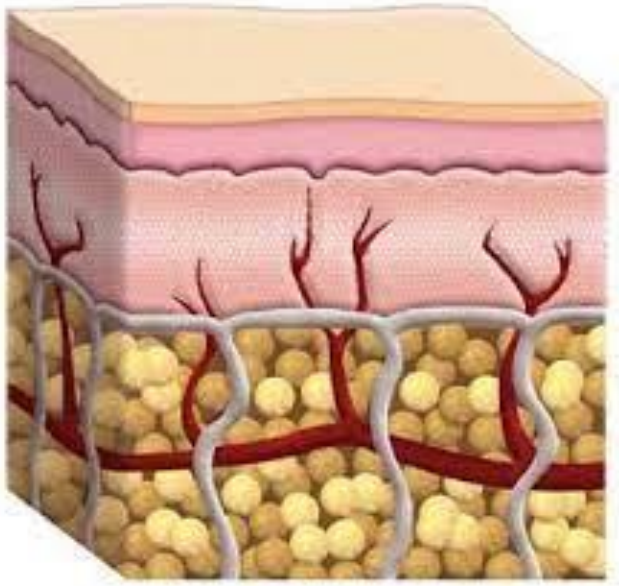
- It is formed by downwards projections of the epidermis (rete ridges) and upwards projections of the dermis (papillae).

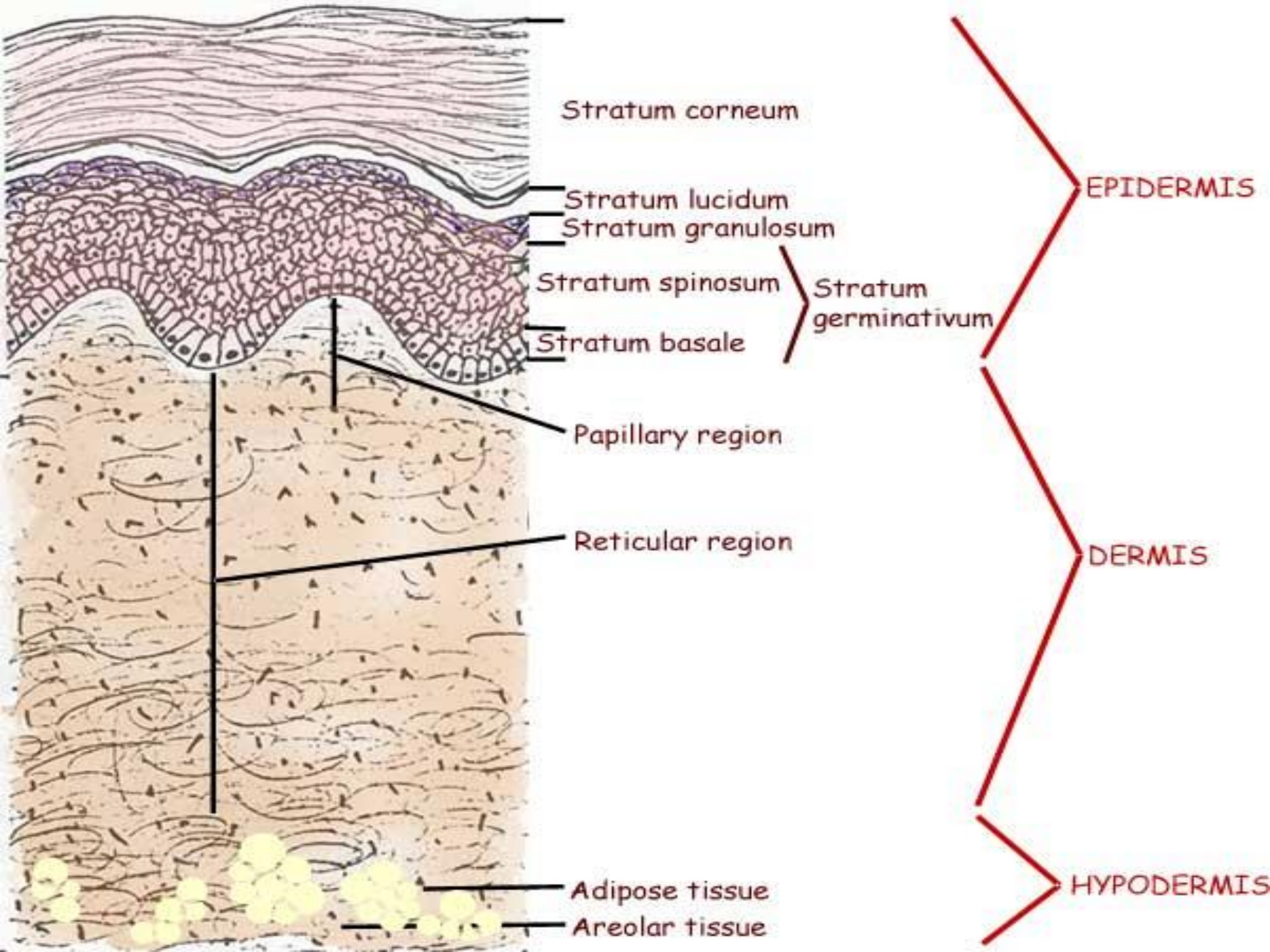
- It is considered to be a semi-permeable filter that permits exchange of cells and fluid between the epidermis and dermis.

- It further serves as a structural support for the epidermis and holds the epidermis and dermis together.

Subcutaneous tissue

- Beneath the dermis lies the subcutaneous tissue which is composed of lobules of fat cells separated by fibrous septa and large blood vessels.




Stratum corneum
Stratum lucidum
Stratum granulosum
Stratum spinosum
Stratum basale
Stratum germinativum
EPIDERMIS
Papillary region
Reticular region
DERMIS
Adipose tissue
Areolar tissue
HYPODERMIS

Functions of the skin

1. It acts as a two-way *barrier* to prevent the inward or outward passage of water and electrolytes. The epidermal barrier is localized to the stratum corneum.

2. The intact stratum corneum *prevents* invasion of the skin by microorganisms.

3. Melanin pigment *protects* the harmful effects of ultraviolet radiation.

4 The skin plays a major role in *thermoregulation* of the human body.

5.The skin plays an important role in the *immunological* host defense.

6. Nerves in the skin are responsible for the *sensations* of touch, pain, warmth, cold and itch.

Basic Skin Lesions

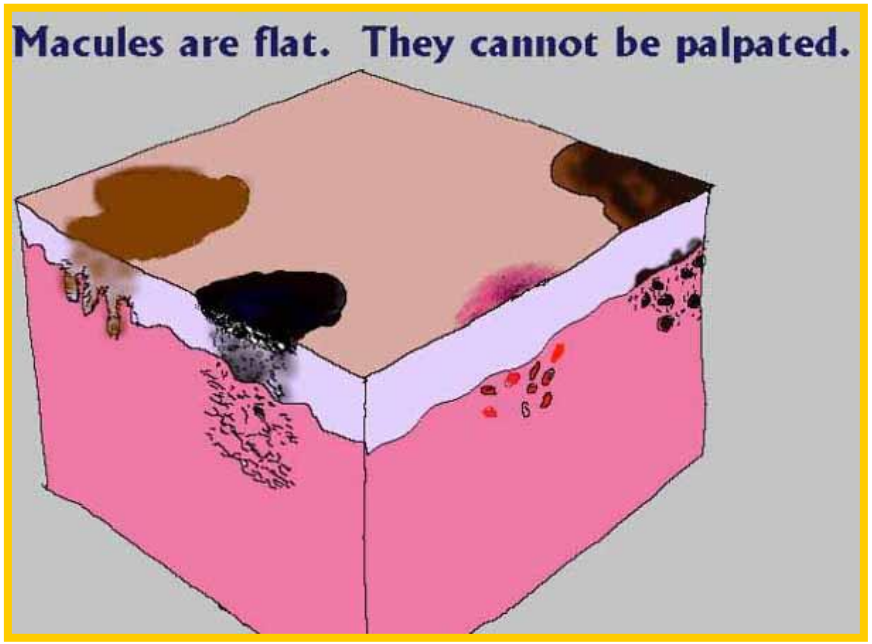
Primary

- Macule
- Patch
- Papule
- Nodule
- Plaque
- Vesicle
- Bulla
- Pustule
- Comedo
- Wheal
- Burrow
- Comedons

Secondary

- Scale
- Crust
- Erosion
- Ulcer
- Fissure
- Excoriation
- Lichenification

Primary lesions:
Macule & Patch





Change in colour or texture of skin.
Circumscribed, flat discoloration.
No elevation.
Less than 1 cm in diameter → macule.
Confluent macules → patch (≥ 1cm).

Primary lesions:

Papule, Nodule,& plaque

An elevated solid lesion

0.5 cm in diameter → papule.

More than 0.5 cm → nodule.

May become confluent → plaques (2 cm or more).

Primary lesions (papule)





Primary lesions (Nodule)





Primary lesions (Plaque)





Primary lesions: Pustule





Circumscribed collection of purulent material (pus). 1ry pustule contains pus from the start. 2ry pustule is a vesicle that was secondarily infected.

Primary lesions: Vesicle & Bulla





Circumscribed collection of free fluid up to 0.5 cm in diameter (vesicle) or more (bulla).

Primary lesions: Wheal



- **Firm edematous plaque resulting from infiltration of the dermis with fluid.**
- **Wheals are erythematous itchy evanescent lesions that may last only a few hours.**
- **Characteristic primary lesion of urticaria**

Primary Lesion: Burrow



- **A small tunnel in the skin, linear, ‘S’ shaped, 3-5 mm in length.**
- **Found in patients with scabies.**

Primary Lesion: Comedons



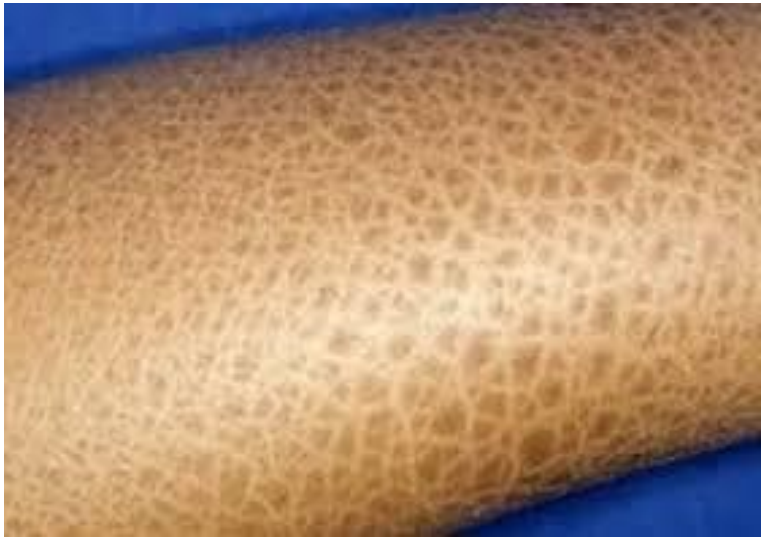
- **Are also called black heads**
- **Found in patients with Acne vulgaris .**

Secondary skin Lesions

- These have developed from modification of primary lesions by regression, infection or trauma.

Secondary lesions: Scale





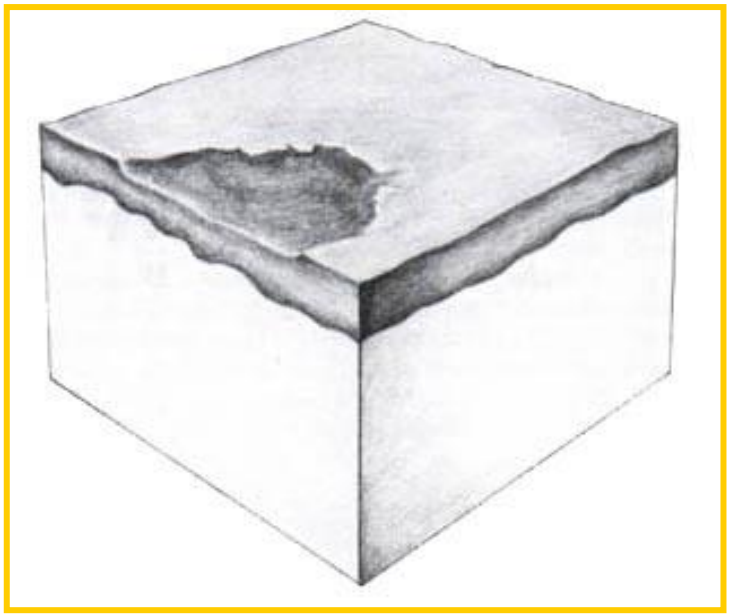
- **Excess keratin produced by abnormal keratinization and shedding.**
- **They may be dry or greasy, large or small.**

Secondary lesions: Crust



Collection of dried secretions

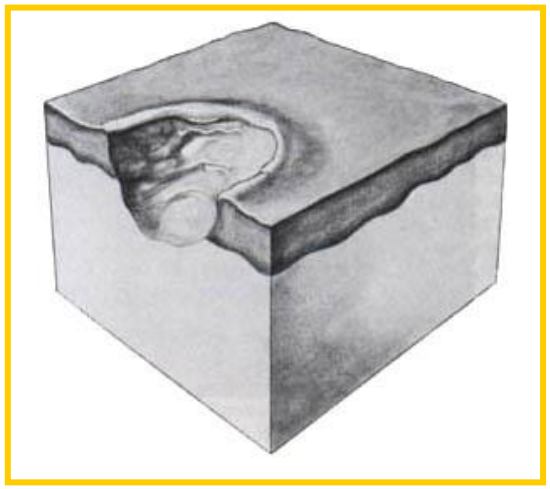
Secondary lesions: Erosion





- **Focal loss of epidermis.**
- **Erosions do not penetrate below the dermo-epidermal junction.**
- **Heal without scarring**

Secondary lesions: Ulcer







- **Focal loss of epidermis and dermis.**
- **Ulcers heal with scarring.**

Secondary lesions: Fissure



Linear loss of epidermis and dermis with sharply defined nearly vertical walls

Secondary lesions: Excoriation





Loss of skin substance caused by scratching with the fingernails to relieve itching.

Secondary lesions: Lichenification



Change in the skin which results from prolonged and repeated irritation. The skin is thickened with accentuation of skin markings and is usually hyperpigmented

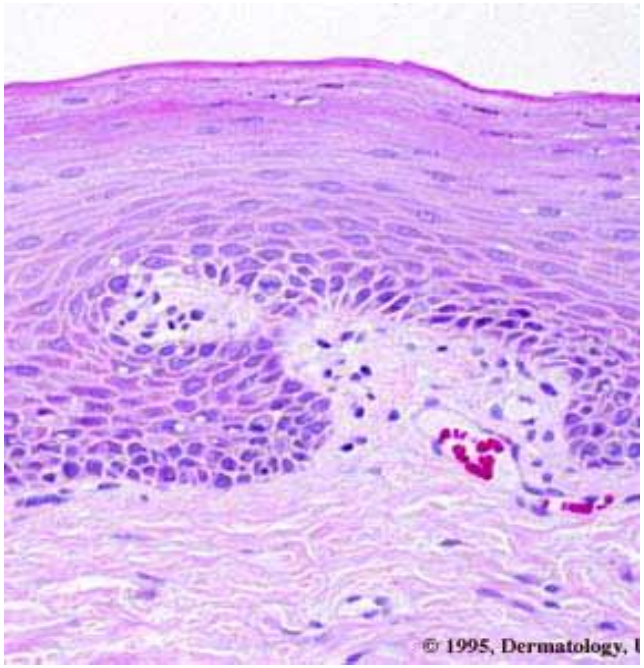
Dermatopathology Terms

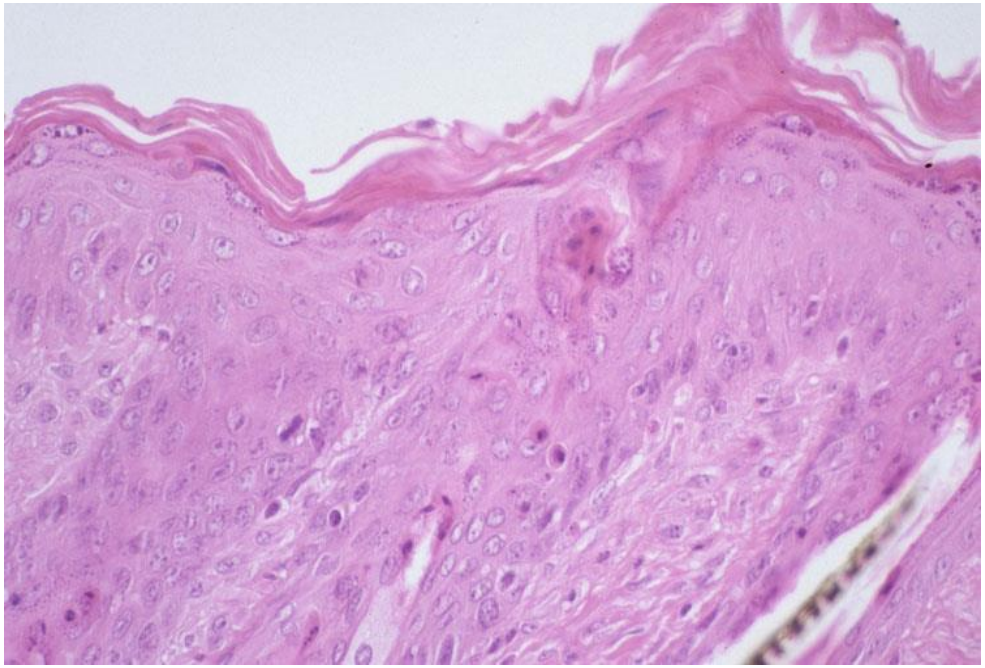
Normal skin



Hyperkeratosis :

Increased thickness of the horny layer.

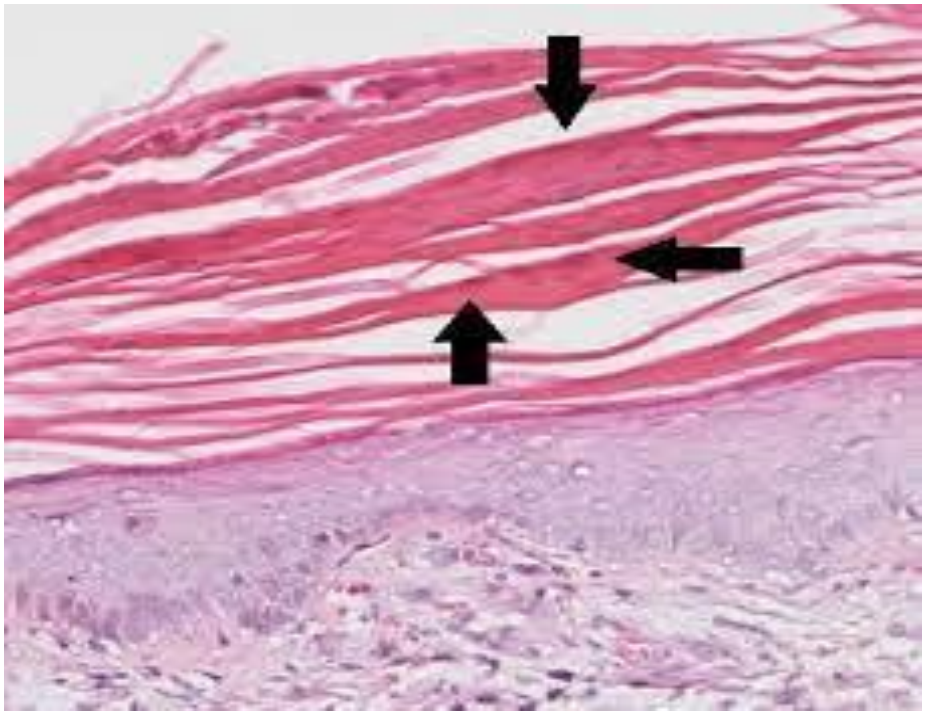




Parakeratosis :

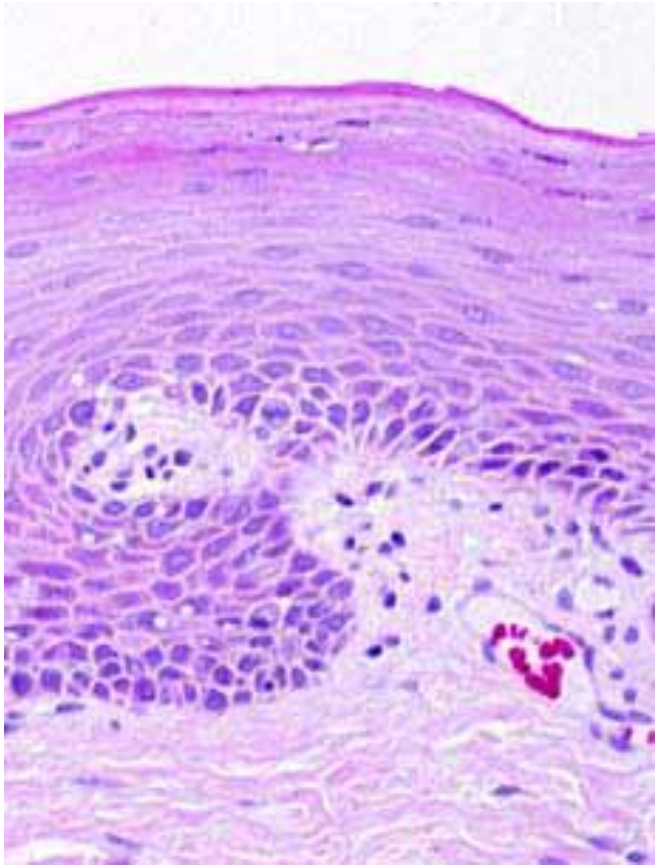
Increased thickness of the horny layer with retention of nuclei due to rapid premature keratinization.

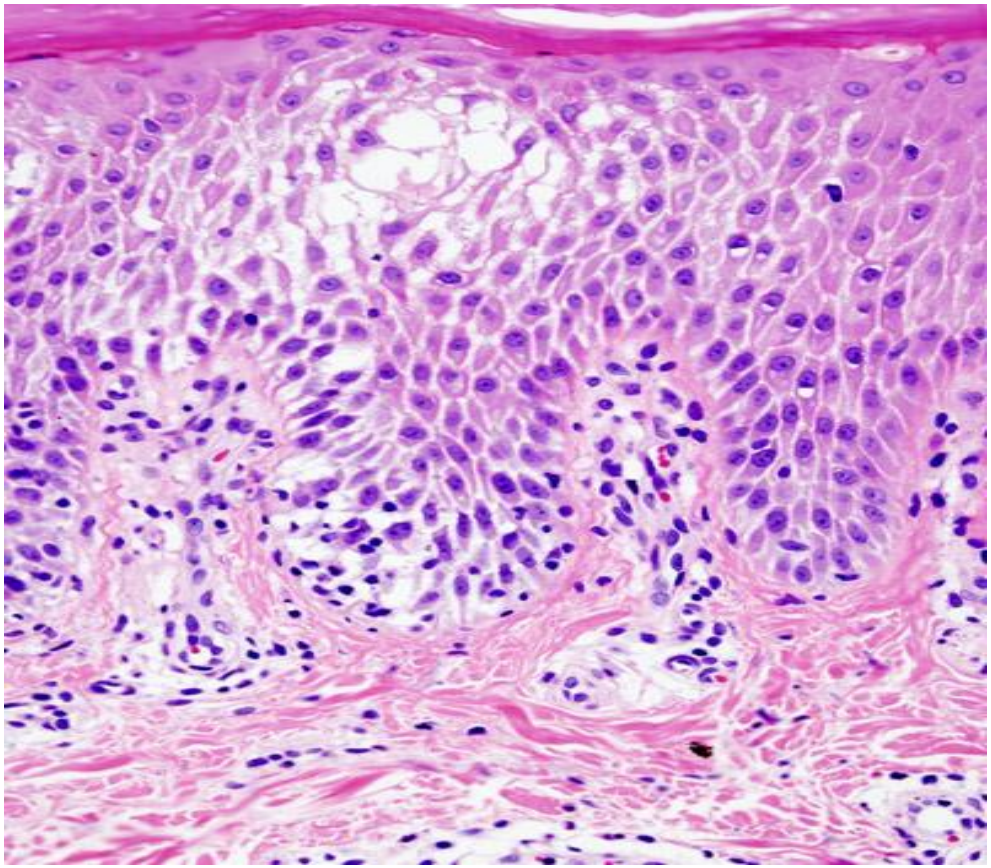




Spongiosis :

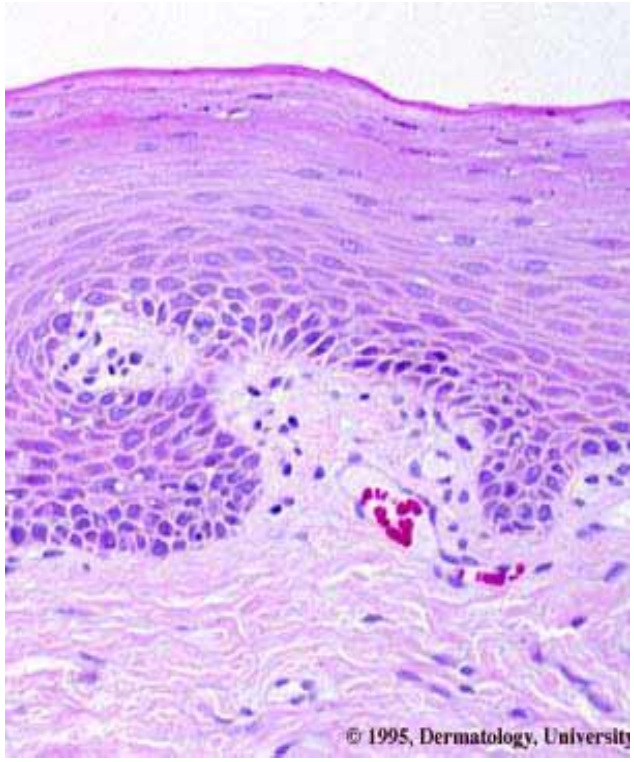
Intercellular edema of the epidermis.

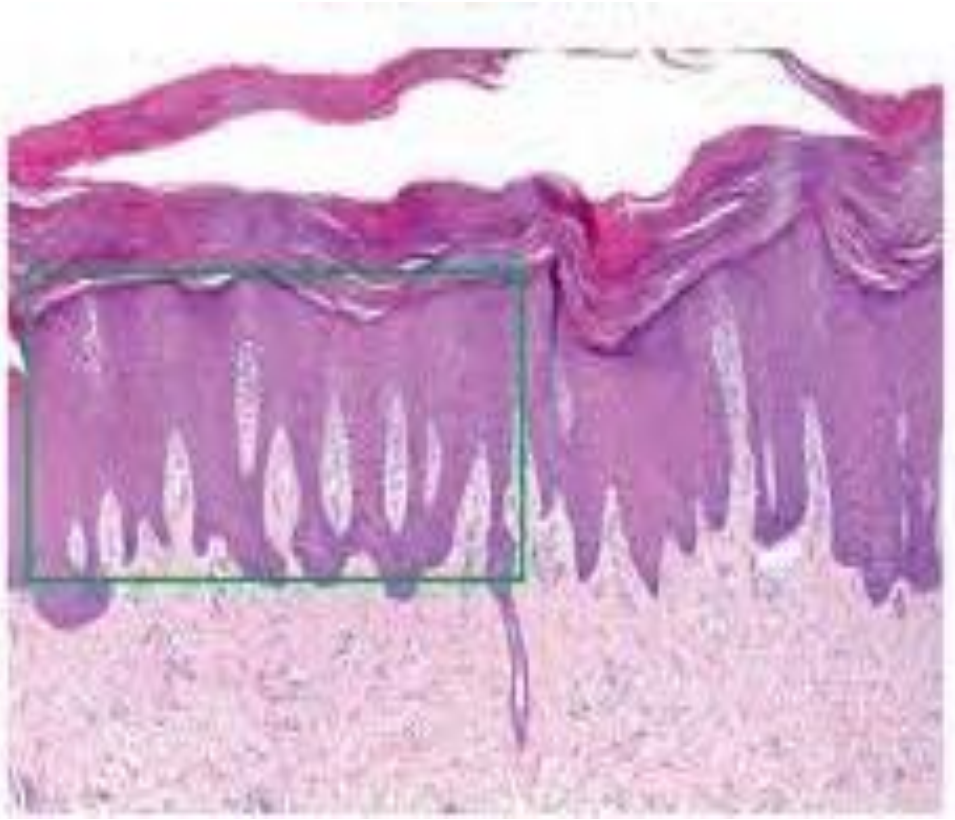




Acanthosis

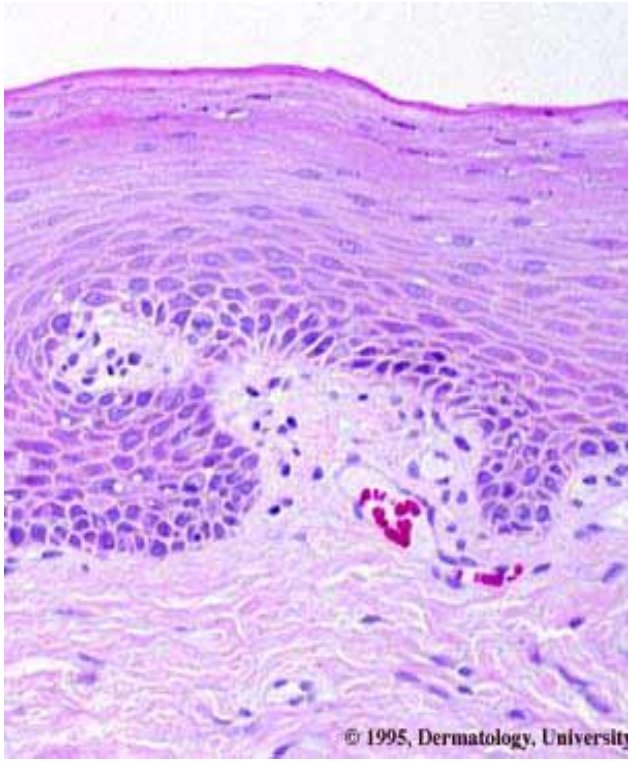
Increased thickness of the prickle cell layer.

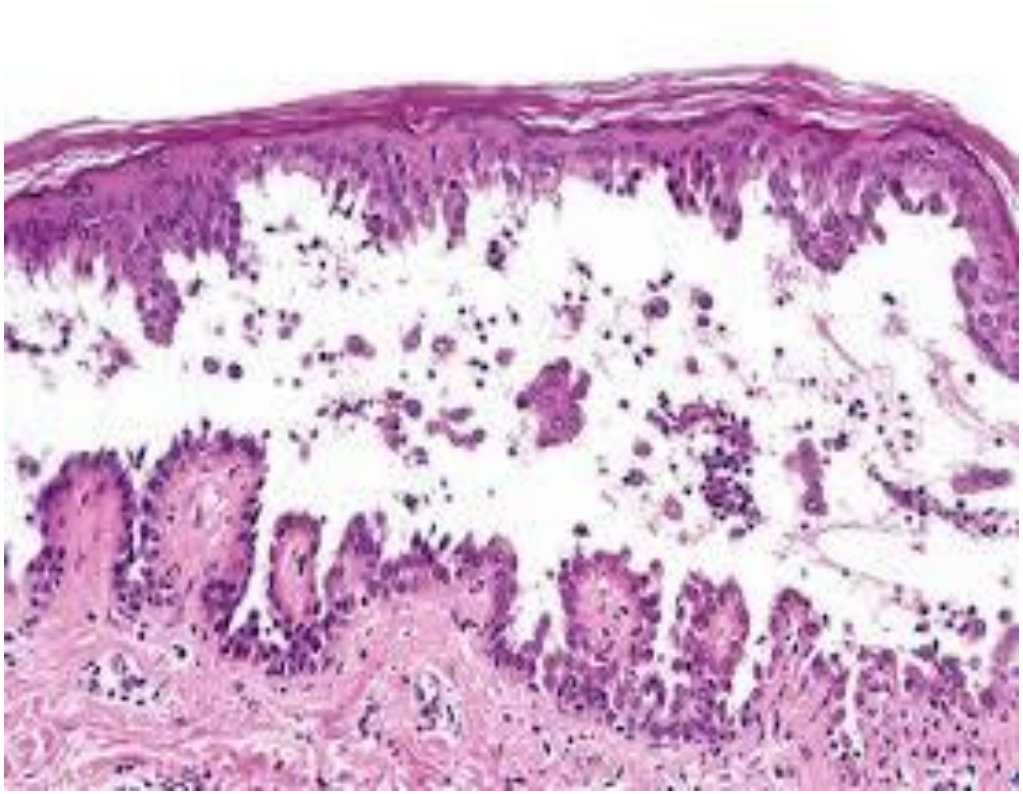




Acantholysis

Loss of cohesion between keratinocytes due to rupture of the intercellular bridges





Thank you

